

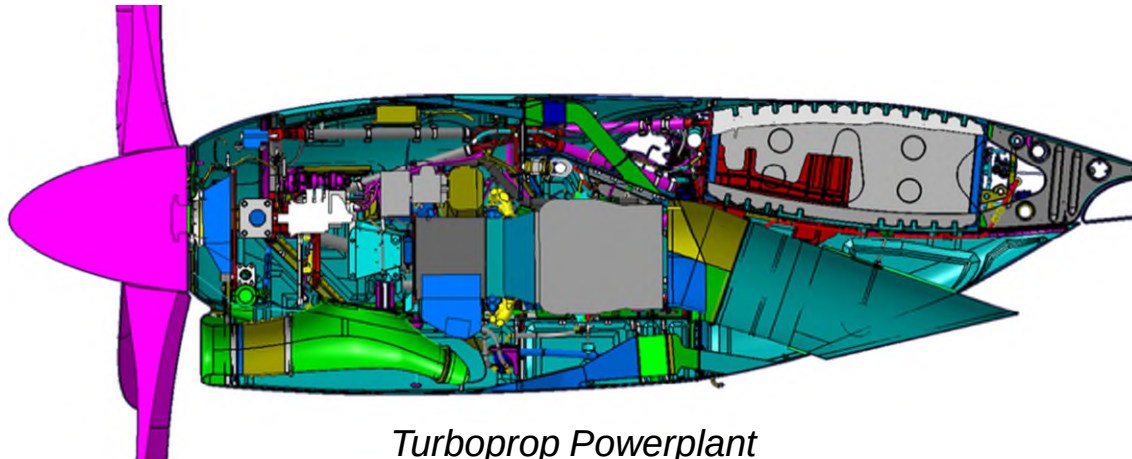
Airframe Technology Master

Design Lecture 33 Engine Mounting System I

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1st Oct 2025

...and what am I supposed to answer at the end?

- Main differences Turbofan vs Turboprop
- Main Load Path in a Powerplant
- Main Certification Sources (i.e. Loads, Fire, Rotor Burst,...)
- Why Isolators in a Turbopropeller installation?
- Why a Truss Structure arrangement?
- Pinned Bars vs Welded Bars
- Pros & Cons Isostatic vs Hyperstatic arrangements
- Induced problems with a “flexible” EMS
- How a Torque Restraint System is acting
- How many bars can I use to support an engine and of which type?



Turboprop Powerplant

L 31

— Functions & Requirements

- Functions
- Certification
- EMS Stiffening
- Maintainability / Accessibility

L 32

— Differences between Turbofan & Turbopropeller Engine Mounting Systems

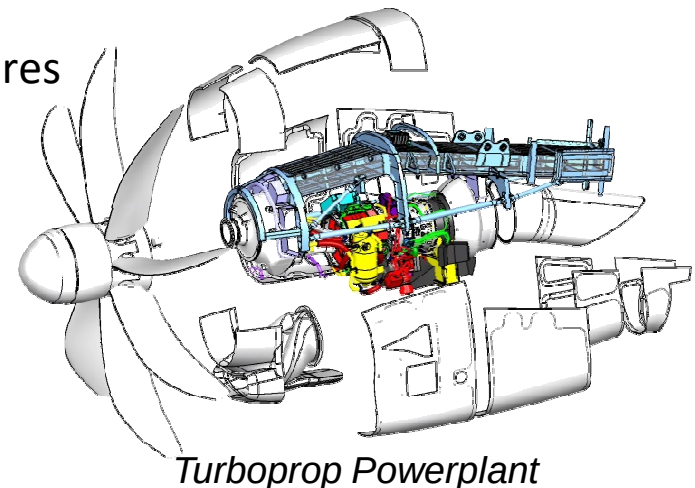
- Engine Structure
- Turbofan EMS typical designs. Why?
- Turbopropeller EMS typical designs. Why?

— Additional Turbopropeller EMS Installation features

- Engine Vibration Isolation System
- Torque Restraint System

— Wing attachments

- Isostatic type
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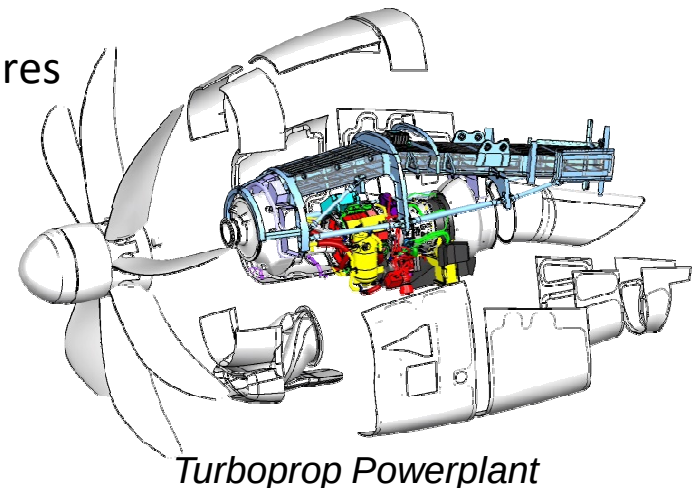
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Functions

Main functions of an EMS are:

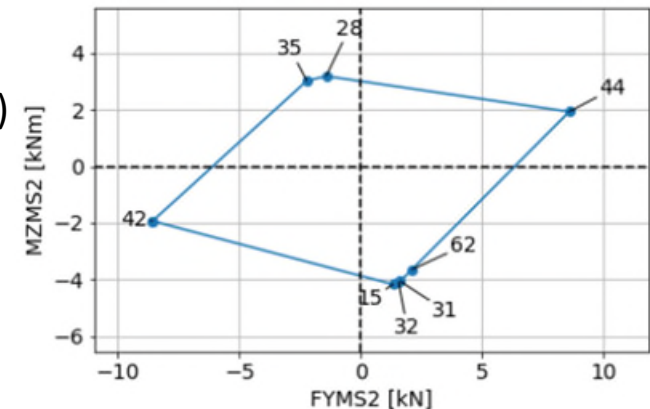
— Engine Loads

- To transfer propeller and engine loads to the rest of the Airframe
- To support the loads that the engine, propeller and the complete powerplant may receive due to A/C behaviour

In general, we can divide the different load cases in different inputs:

- Certification bookcases (See Certification Requirements below)
- “Static” loads
 - Due to flight manoeuvres
 - Due to Propeller 1P loads (Aerodynamic loads)
- Dynamic loads
 - Gusts
 - Ground (Dynamic Landing Loads)
- Specific Engine loads
 - Thrust
 - Reverse thrust
 - Torque
 - Gyroscopic effects

2D Diagram example



All these loads are usually calculated in a dedicated Powerplant loads set and given in specific monitor stations (mount planes, Propeller to Engine interface, CoG,...) and presented in 2D diagrams as requested by the Powerplant Stress engineers

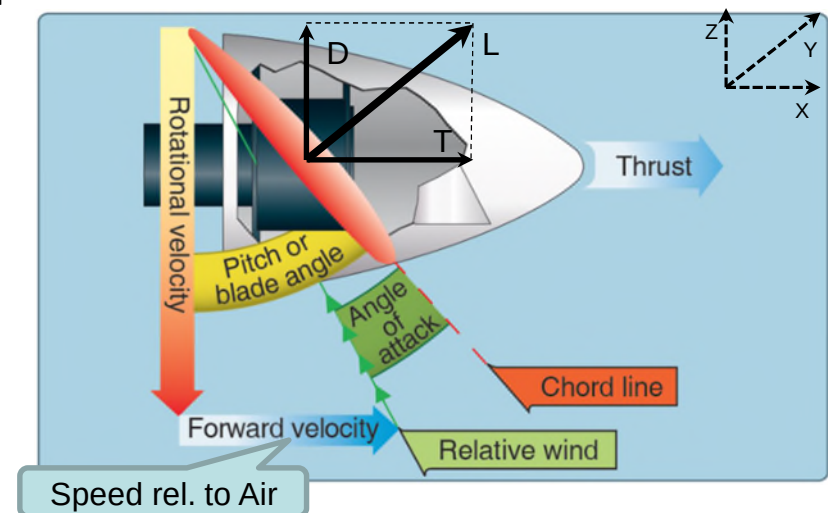
Functions

Main functions of an EMS are:

— Engine Loads (cont'd)

▪ 1P Loads

- Special mention requires the so called 1P Loads in propelled aircraft, or more generally nP Loads.
- Those are produced due to the relative angle between the in flow airstream and the propeller rotating axis
- If that angle is not 0° , the blade will not see the same angle of attack all along the 360° path
- This produces loads and moments in the Propeller plane (F_z , F_y , M_z , M_y)
- These loads have a frequency of 1P (once per blade cycle) and their harmonics (nP)



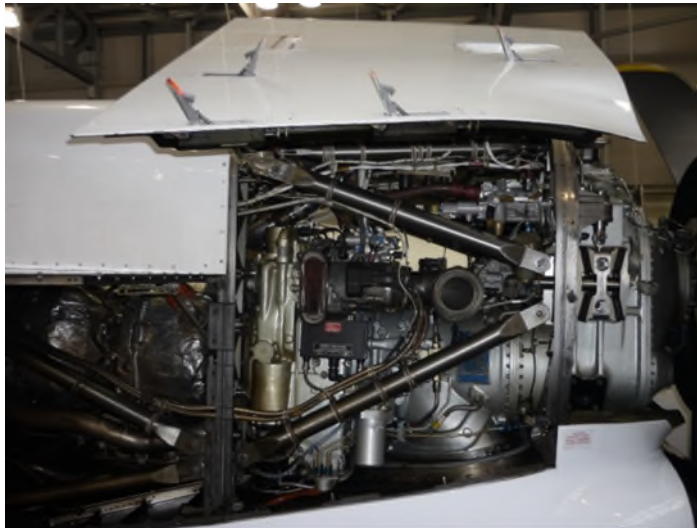
Air to Propeller speeds

Functions

Main functions of an EMS are (cont'd):

— Cowling Loads

- The Cowls are supported by the EMS, thus they will receive the aerodynamic loads and will transfer them to the EMS
- In some applications, the cowls are also required to be part of the main load structural path, usually as a backup structure in case of failure of one EMS element (bars,...)



Bombardier DASH 8 Nacelle

- Additionally, maintainability devices such as Hold Open Rods, Actuators and similar features required to operate the Cowls may transfer also loads to the EMS, usually on ground only, together with the so called “monkey loads” when required

Functions

— Maintainability Allowances

The engine, propeller and its associated systems and equipment within the powerplant will require a frequent access for maintainability, therefore the EMS architecture will have as one the most important requirements, the maintainability that will drive the specific designs we may found in the powerplant structure as compared to other aircraft structures.

- Engine removal and installation

Nevertheless, the maintenance requirement that will drive more deeply the EMS architecture will be the Engine installation and removal operation.



Engine Removal Operation

Functions

— Others

- Powerplant Systems support

The EMS will be also the support for the complete powerplant system package. Among them, we can find some relevant elements like different radiators for different purposes, Engine and/or Propeller computers and minor components like piping, harnesses, valves, etc.

During the EMS design we should consider as much as we can all those elements giving allowance for them while keeping the main structural path as clean as possible.

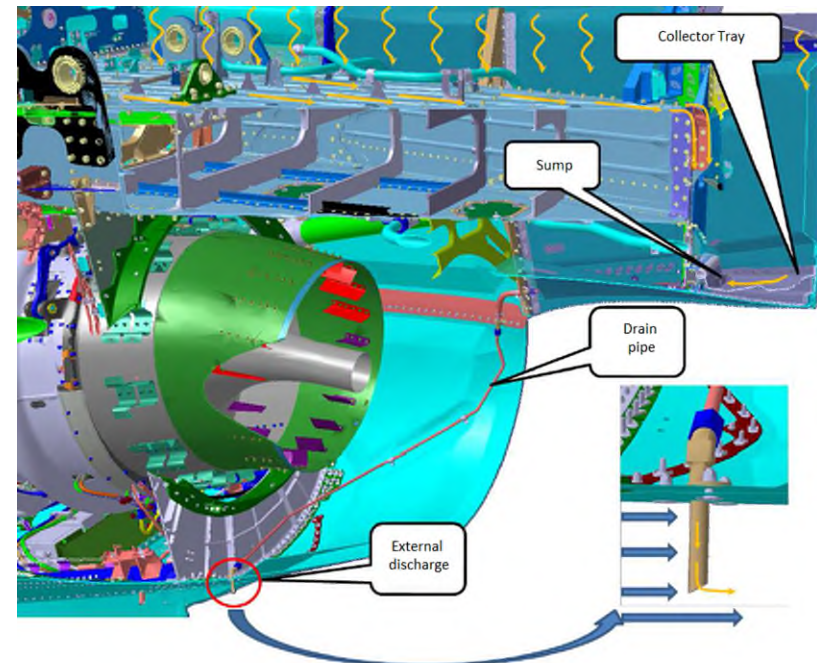
In general terms we will avoid system installation (other than small pipes and harnesses) on the middle of bars to reduce buckling and/or damages to them as most of the system components are prone to be removed and reinstalled during the whole aircraft life several times.

- Ventilation & Drainage

As part of the powerplant ventilation and drainage system, we should also consider during the EMS design the air path through the Engine Nacelle trying to avoid air blocking and flammable fluids being trapped locally in a quantity bigger than a small quantity (usually around 44 cm^3). After a test more than 90% must be recovered.

- Accessibility for inspection

Man holes might be required depending on the design.



EMS Drainage

Certification Requirements

To note that the following requirements does not apply to the engine itself, that it is covered by CS-E, FAR-E or similar. In a similar way, in turboprops, propeller certification requirements are covered by CS-P, FAR-P or similar.

Therefore, the following requirements may apply to, structures, systems and their integration into the aircraft, including the engine and propeller installation as part of the powerplant.

— Powerplant requirements

- CS 25.901 Installation
- CS 25.1309 Equipment, Systems and Installations

— Loads

- CS 25.341 Gust and turbulence loads
- CS 25.361 Engine and auxiliary power unit torque
- CS 25.362 Engine failure loads
- CS 25.363 Side load on engine and auxiliary power unit mounts
- CS 25.571 Damage-tolerance and fatigue evaluation of structure
- CS 25.631 Bird strike damage
- Other structural paragraphs like CS25.302 (Loads), CS25.303 (Factor of Safety), CS25.305 ((a)(b) Strength and deformation) & CS25.307 (Proof of structure) shall be also used

To start with, as a preliminary design, engine torque and seizure loads are used meanwhile the complete loads pack is produced.

Certification Requirements

— Fire protection

- CS 25.865 Fire protection of flight controls, engine mounts, and other flight structure
- CS 25.1191 Firewalls
 - Fireproof
 - Fluids leakage
 - Sealing
 - Corrosion
- CS 25.1193 Cowling and Nacelle skin (See Cowlings lectures)

— Uncontained Engine Rotor Failure / Torching Flame

In the UERF event, the probability of having a catastrophic event, shall never be higher than 1/20 (5%) assuming that the event probability itself is 1 (100%).

This requirement may have a big impact mainly in engine and other systems location in the aircraft as well as to the EMS definition, nevertheless, the regulation requires an implementation of a **practical design** (i. e.: usually engine location will not have an asymmetrical location due to this requirement).

Therefore, the EMS definition will maintain the failed engine in place after such an event unless we can demonstrate that, after the event, the engine will have a “safe separation” without producing damages to the aircraft which is difficult to justify in case of propellers.

Certification Requirements

■ UERF

- CS-E (Engine Certification) (Does not apply to aircraft manufacturer design)
- CS-P (Propeller Certification) (Does not apply to aircraft manufacturer design)
- CS 25.903(d)(1) Engines. Turbine engine installations
 - (See AMC 25.903(d)(1) and AMC 20-128A)
- CS 25.905(d) Propellers
 - (See AMC 25.905(d))
- CS 25.1193 Cowling and nacelle skin



Turbine rotor burst on a Qantas Flight 32
(A380-842 MSN014) Nov 2010

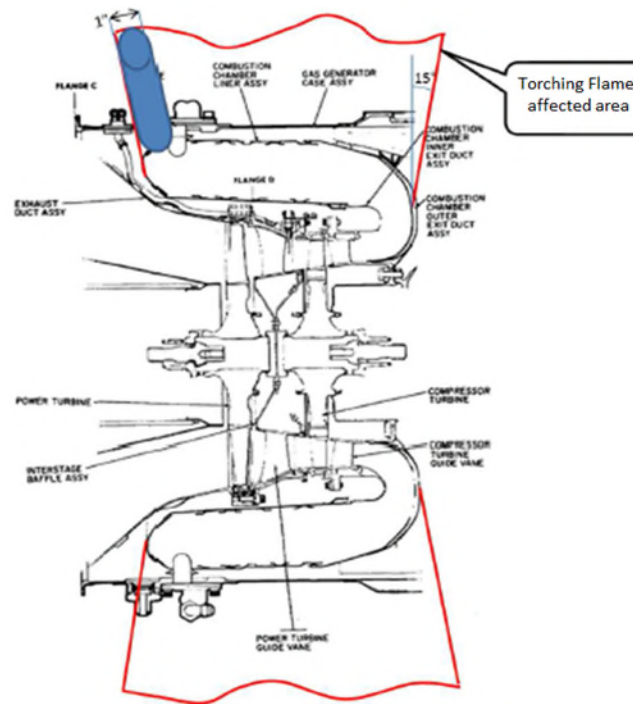


Image source: Courtesy of the 'Posmetro' newspaper, Indonesia.



Certification Requirements

- **Torching Flame** (Combustion Chamber burn through)
 - CS 25.903(d)(1) Engines. Turbine engine installations
 - (See AMC 25.903 (d)(1) → BS 3G 100-2.3.13:1973)



Torching Flame affected area definition

Certification Requirements

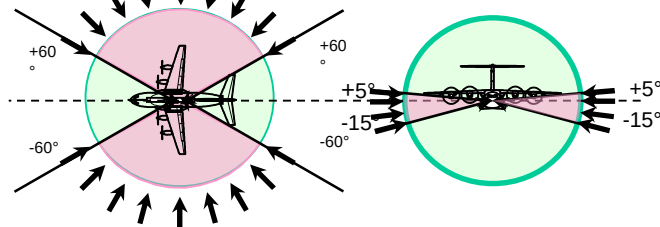
— Propeller clearance

Those clearances applies not only to the propeller blades, but to any rotating part (hub, spinner,...), in this sense, the EMS front part and isolators mainly may have low clearances to the rotating parts.

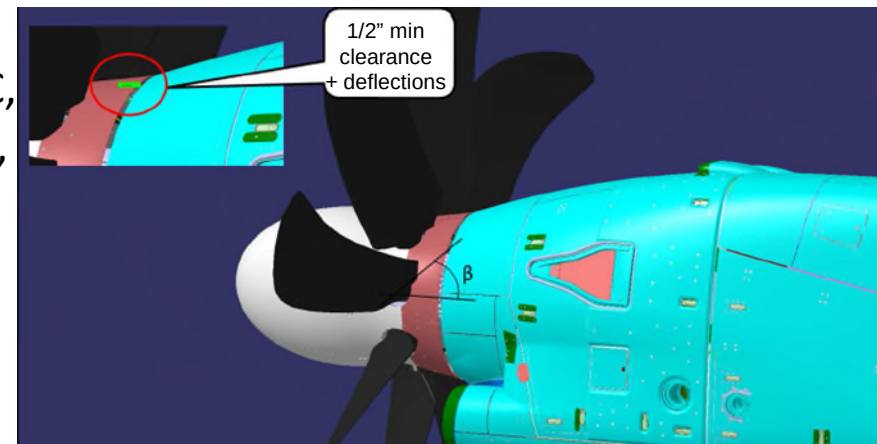
- CS 25.925 Propeller clearance (c) (2)(3) Structural clearance
 - (2) At least 13 mm (0.5 inches) longitudinal clearance between propeller blades or cuffs and stationary parts of the aeroplane; and
 - (3) Positive clearance between other rotating parts of the propeller or spinner and stationary parts of the aeroplane.

— Battle Damage (Only for military A/C)

Specific requirements may apply to every A/C, including bullet size and speed, on top of this, Def Stan 00-970 might be used as related to bullet directions.



Trajectories for Battle Damage

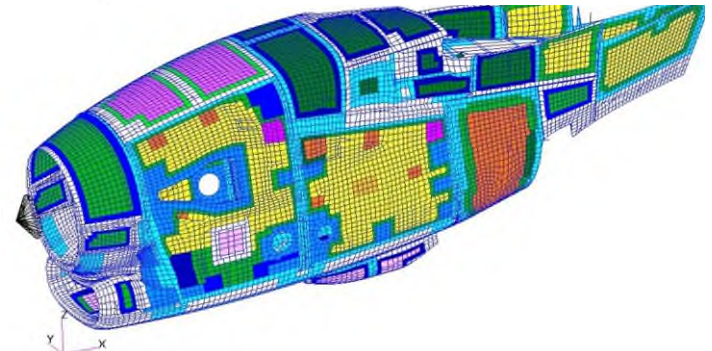
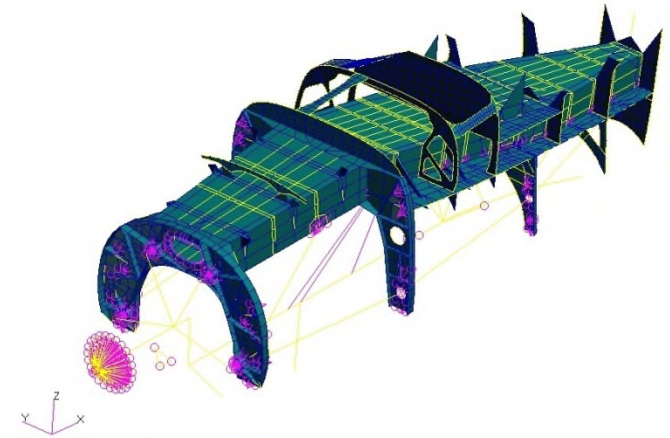
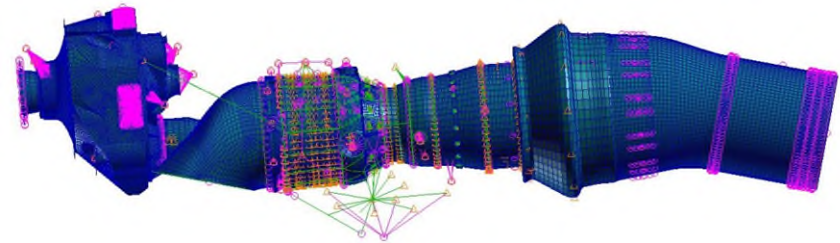


Minimum distances

EMS Stiffening

— EMS stiffness contribution to Cowls loading and Engine clearance control

- Three components must be considered when talking about powerplant stiffness, EMS, Engine and Cowls.
- If EMS contribution is relatively low compared to the Cowls, these will be highly loaded even on ground. Then, the maintainability can be compromised due to complex operation (i.e. many latches, adjustability difficult to maintain,...)
- The Engine contribution must be also under control to avoid the engine tip clearances to be compromised, thus reducing engine life and increasing consumption
- An Integrated Finite Element Model is highly recommended to fine tune the different components from the very early design phase



Powerplant Integrated FEM

Maintainability / Accessibility

— Engine accessibility “On wing”

As mentioned before, the engine, propeller and its associated systems and equipment within the powerplant will require a frequent access for maintainability, more specifically:

▪ Accessibility to equipment and components

The engine supplier will, generally, specified two different frequency access levels to the different engine equipment and to the engine core for borescope inspections, usually equipment with:

→ **Frequent access** would be:

- Permanent Magnet Alternator (Engine own alternator)
- Fuel valves, Anti-Ice valves, etc...
- Borescope access ports
- Engine gearbox equipment (Fuel Pump, Oil Pump,...) and propeller equipment in propeller gearbox (Aux. feathering Pump, Prop Oil Pump, Beta angle selector,...)
- Hydraulic Pump
- A/C Electrical Generator
- Fuel Control Unit
- Pneumatic or electric Starter and its shut off valve

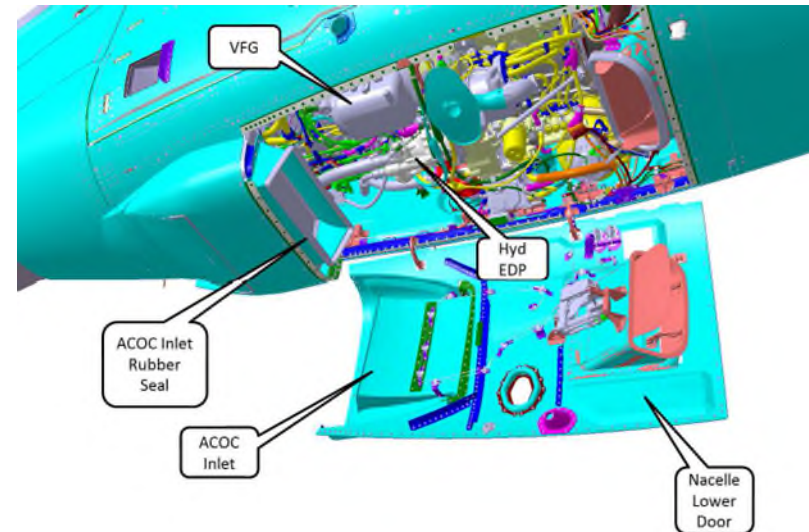
Maintainability / Accessibility

→ ...and with **Very Frequent access** would be:

- Fuel Nozzles
- Engine and Propeller various sensors (vibration, pressure, temperature,...)
- Oil Filters and its By-Pass indicators
- Chip Detectors
- Oil filler Cap
- Oil level indicator
- Thermocouples
- Ignitors

The engine supplier will also specified the required space to access those elements, and these requirements shall always be considered within the early design phases of the EMS

The incorporation of Supportability teams are also key in this phases.



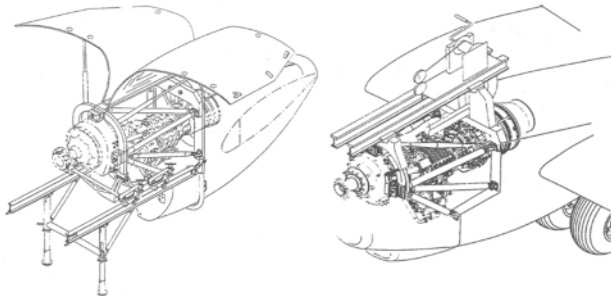
Accessibility "On wing"

Maintainability / Accessibility

— Engine removal and installation

Engine removal and installation will, generally, lead to an EMS design opened at the bottom in order to allow an easy engine removal and, thus, forcing us as designers, to a non-structural efficient design in the case of a turbopropeller installation for example. This will usually load the lower EMS elements much more than the upper ones.

There are design applications that solve that problem by closing the structure at the bottom either by using dedicated transversal links or using lower cowls as main load path, and therefore behave as part of the EMS. This design will make maintenance much more complex as those bottom added structural elements will always carry some loads, even in ground, due to engine and propeller weight and this might force to hang the engine to compensate the engine weight or design complex attachments for those links or cowls.



Two different concepts for a turboprop engine removal



Lower Cowl load path concept

Maintainability / Accessibility

— EMS accessibility

The EMS, as a structure, requires accessibility to its different components and it will require mainly two different elements to be considered during the design phase:

- **EMS box (Pylon type):** In this case, the box will require hand holes to access to every single area, the size and amount will be as required to the specific design
- **EMS truss structure type:** In this case, although we can find several types of truss structures, the major impact can be on welded arrangement where the design will be on the contrary, it is, to avoid the need of access by closing completely the assembly from the outside (See Manufacturing methods). By any design method we should avoid removal of structural elements (i.e.: pinned bars) to access any powerplant equipment.

